Copy-and-replace build sheet — GT Designer3

The ten screens contain 151 objects, but most of them are the same handful of objects repeated six times. Build **one** of each, copy it, then change only the device numbers. That turns the job into roughly 40 operations.

The technique

1. Build the first instance completely — style, text, colours, trigger, everything.
2. Select the whole group and **copy-paste** it five times, positioning by the design PDF.
3. Retarget the copies. Two ways, both far faster than opening each dialogue:
    - **Data Browser** (`View → Docking Window → Data Browser`) lists every object on the screen with its device in a grid. Edit the device cells directly.
    - **Device replace** (`Search/Replace → Device Replace`), scoped to the selected objects, one substitution at a time.
4. Work down the substitution table for that screen. Tick each row.

> *Do the copy **before** you set the triggers and gating, not after — a wrong trigger copied six times is six things to fix.*

Screen 1100 — cure slot tiles

Build the **SLOT 1** tile in full, then copy it five times.

One tile contains six objects:

Object	Recipe	Device on slot 1
State chip	R2 Word Lamp	D70
Elapsed hours	R7/R11	D40
Elapsed minutes	R7/R11	D46
Elapsed seconds	R7/R11	D52
Progress bar	R10 Level	D64
Remaining minutes	R7	D58
START switch	R3 Set	M800
RESET switch	R3 Set	M810
RESET enable condition	Operation Condition	M4008

Then replace, per copy:

Copy	State	Hours	Mins	Secs	Bar	Remain	START	RESET	RESET enable
SLOT 2	D71	D41	D47	D53	D65	D59	M801	M811	M4009
SLOT 3	D72	D42	D48	D54	D66	D60	M802	M812	M4010
SLOT 4	D73	D43	D49	D55	D67	D61	M803	M813	M4011
SLOT 5	D74	D44	D50	D56	D68	D62	M804	M814	M4012
SLOT 6	D75	D45	D51	D57	D69	D63	M805	M815	M4013

Every device steps by exactly one per slot, so if a copy is out by one it will be out by one on all six columns — easy to spot in the Data Browser grid.

Screen 1000 — the six mini-tiles

Smaller: state chip, three-part time, progress bar. No switches.

Copy	State	Hours	Mins	Secs	Bar
SLOT 1 (build this one)	D70	D40	D46	D52	D64
SLOT 2	D71	D41	D47	D53	D65
SLOT 3	D72	D42	D48	D54	D66
SLOT 4	D73	D43	D49	D55	D67
SLOT 5	D74	D44	D50	D56	D68
SLOT 6	D75	D45	D51	D57	D69

The rest of 1000 is built once each: four status lamps (`M32`, `M33`, `M35`, `M10`), four switches (`M820–M823`), the PID lamp (`M11`), the door timer (`D18`), and three summary cells (`D24` 32-bit, `D11`, and the alarm count).

Screen 1700 — the six early-reset rows

One row: slot label, three-part elapsed time, remaining minutes, and an EARLY RESET switch with a confirmation dialogue.

Copy	Hours	Mins	Secs	Remain	EARLY RESET switch	Row visible when
SLOT 1 (build this one)	`D40`	`D46`	`D52`	`D58`	`M810`	`D70` = 1
SLOT 2	`D41`	`D47`	`D53`	`D59`	`M811`	`D71` = 1
SLOT 3	`D42`	`D48`	`D54`	`D60`	`M812`	`D72` = 1
SLOT 4	`D43`	`D49`	`D55`	`D61`	`M813`	`D73` = 1
SLOT 5	`D44`	`D50`	`D56`	`D62`	`M814`	`D74` = 1
SLOT 6	`D45`	`D51`	`D57`	`D63`	`M815`	`D75` = 1

The switch device is the **same** `M810`–`M815` used for the normal reset on

1100. The PLC decides which rule applies from whether the slot is complete, so there is no second set of bits to wire.

Whole screen gated on `M58`. Confirmation dialogue on all six.

Screen 1600 — the keypad

Twelve keys. Build **one** digit key, copy it eleven times, change only the text and the value written.

All twelve write to the GOT's own numeric input buffer, not to a PLC device — only the two login buttons touch the PLC. So the copies need no device change at all, just the caption.

Key	Caption
1	`1`
2	`2`
3	`3`
4	`4`
5	`5`
6	`6`
7	`7`
8	`8`
9	`9`
CLR	`CLR`
0	`0`
DEL	`DEL`

Screen 1500 — the nine settings rows

Build **one** row — label text, Numerical Input, unit text, range text — then copy eight times and change the device, the range and the two texts.

Row	Device	Range	Unit
Heat-up watchdog	`D4100`	10 – 240	min
Door watchdog	`D4101`	60 – 1800	s
Chatter transitions	`D4102`	4 – 50	—
Chatter window	`D4103`	5 – 60	s
Heater minimum ON	`D4104`	5 – 120	s
Heater minimum OFF	`D4105`	5 – 120	s
Manual test timeout	`D4106`	60 – 900	s
Auto-logout	`D4107`	60 – 900	s
Cure time	`D4108`	fixed 7200	s

`D4108` is **32-bit** (recipe R8). The other eight are 16-bit.

All nine gated on `M58`. Each also needs the `M831` activity ping as a second action.

Built once, not copied

Everything else is a one-off. Roughly forty objects in total:

Screen	Objects
Header / footer template	1 rectangle, 2 texts, 1 word lamp, date, time, 6 nav switches
1200 Counters	6 × 32-bit numerics, 3 × 16-bit, 1 alarm-driven panel, 1 gated switch, 3 recent-reset texts
1300 Alarms	1 alarm display, 1 accept switch, 1 nav switch, 1 count
1400 Manual test	3 momentary switches, 3 feedback lamps, 5 condition lamps, 1 numeric, 1 exit switch
1600 Login	1 masked input, 2 login buttons, 1 lockout lamp, 1 countdown, 1 logout switch
1800 Counter reset	5 numerics, 2 switches with dialogues
1900 History	1 history display, 1 USB export switch

Checking it afterwards

Open the **Data Browser** with all screens shown. It lists every object and its device. Compare that against `device-crossref.csv`, which lists the same 64 devices and where each should appear.

The three things worth checking specifically, because they fail quietly:

Check	Why
Every device in the 32-bit list is set to 32 bit	reads fine to 32767, then goes negative
`M800`–`M815` and `M820`–`M830` are **Set**, not Alternate or Momentary	a command could repeat or be missed
`M832`–`M834` are **Momentary**	otherwise a manual-test blower will not stop

32-bit devices: `D24`, `D4000`–`D4011`, `D4020`, `D4022`, `D4024`, `D4026`, `D4028`, `D4030`, `D4032`, `D4034`, `D4108`. Everything else is 16-bit.